

AWS DevOps Engineer Intern Assignment — Documentation Report

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GitHub Repo: <https://github.com/MantashaR/aws-devops-assignment> **EC2 Public IP:** 3.109.60.228

1. Objective

Deploy a static website on an AWS EC2 instance running Ubuntu, served by Nginx, and document the end-to-end process.

2. AWS Services Used

Service	Purpose
EC2	Virtual server (Ubuntu 24.04 LTS, t3.micro free tier)
Security Groups	Virtual firewall — opened ports 22 (SSH) and 80 (HTTP)
Key Pair	RSA key (<code>devops-key.pem</code>) for secure SSH authentication

3. Linux Commands Used

Command	Purpose
<code>sudo apt update && sudo apt upgrade -y</code>	Update & upgrade packages
<code>sudo apt install nginx -y</code>	Install Nginx web server
<code>sudo systemctl status nginx</code>	Check Nginx service status
<code>sudo systemctl restart nginx</code>	Restart Nginx
<code>df -h</code>	Check disk usage
<code>free -h</code>	Check memory usage
<code>ps aux --sort=-%mem head</code>	List top running processes
<code>scp -i key.pem index.html ...</code>	Copy website file to instance
<code>sudo cp ... /var/www/html/</code>	Replace default Nginx page

4. Steps Performed

1. Launched an Ubuntu 24.04 t3.micro EC2 instance (ap-south-1 / Mumbai).
2. Created a Security Group allowing SSH (22) and HTTP (80).

3. Connected to the instance over SSH using the downloaded key pair.
4. Updated packages and installed Nginx; verified it was running.
5. Checked disk, memory, and process usage (Linux basics).
6. Uploaded a custom `index.html` and replaced the default Nginx page.
7. Verified the site loads at the EC2 public IP in a browser.
8. Pushed all files + README to GitHub.
9. (Bonus) Wrote and ran a Nginx restart shell script (`restart-nginx.sh`).

5. Problems Faced & Solutions

Problem	Solution
SSH "Permission denied (publickey)"	Ran <code>chmod 400 key.pem</code> ; used <code>ubuntu@</code> as the user
Website not loading in browser	Port 80 was missing from the Security Group — added it
Default Nginx page still showing after copy	Restarted Nginx after replacing <code>/var/www/html/index.html</code>
<code>.pem</code> too-open-permissions warning on SSH	Fixed key file permissions to <code>400</code>

6. Learnings

- How a Security Group acts as a stateful virtual firewall at the instance level.
- The difference between a dynamic public IP and a static Elastic IP.
- Nginx serves from `/var/www/html/` by default; how `systemctl` manages services.
- Basic Linux server administration: package management, service control, resource monitoring.
- Using SSH key pairs and `scp` for secure remote access and file transfer.

7. Total Time Taken

Approx. 1.5 hours — EC2 setup (~20 min), Nginx + deployment (~20 min), bonus script (~15 min), Git & docs (~35 min).

Deliverables: GitHub repo link, this report (as PDF), EC2 public IP, and screenshots — bundled in a ZIP.